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Insight: Financial Report Analysis Engine based on LlamaIndex

CS6493 NLP Final Project

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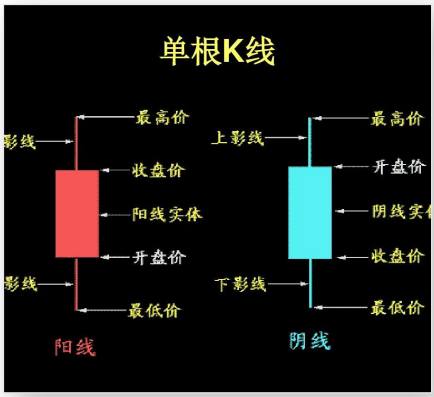
01. The Challenges of Financial Analysis



- Information Overload & Effi ciency Bottlenecks
- High Demand for Timeliness
- Data Privacy & Security

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01. Why RAG over Fine-tuning?

- Knowledge Update Instantaneity
- Cost-Effectiveness
- Reduction of Hallucination

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01. Tech Stack: The Engine Room

Core Framework

- LlamaIndex v0.10+

Foundation for orchestrating data pipelines and connecting to LLMs.

Vector Database

- ChromaDB

Lightweight, in-memory DB for storing and retrieving vector embeddings efficiently.

AI Models

- Qwen2.5-7B (Gen) / 72B (Eval)
- Embedding: BGE-M3
- Reranker: BGE-Reranker

Pipeline & Interface

- Frontend: Gradio
- PDF Parsing: PyMuPDF
- Evaluation: RAGAS

Hardware Infrastructure

- Apple Mac Studio M3 Ultra

512GB RAM & MPS Acceleration for fast local inference.

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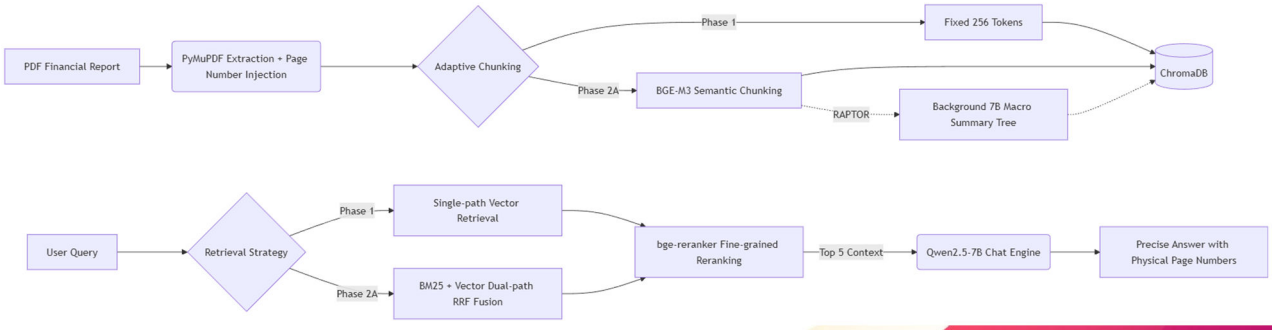
02. System Architecture Overview



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02. Data Ingestion & Retrieval

Data Ingestion Optimization

- **Metadata Contract:** Enforce **source** and **page_label** for full traceability of every generated response.
- **Chunking Strategy:** Utilize semantic chunking (BGE-M3) to preserve logical coherence, which is critical for causal reasoning tasks.
- **RAPTOR Index:** Construct a hierarchical knowledge structure to support efficient macro summarization across large documents.

Retrieval Strategy Optimization

- **Hybrid Retrieval:** Combine the strengths of **vector search** (capturing semantic similarity) and **BM25** (lexical keyword matching) to achieve more robust and comprehensive recall.
- **Engineering Optimization:** Implement a global singleton cache for the BM25 inverted index matrix and Reranker model, enabling zero-latency reuse and significantly faster overall system response times.

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02. Generation Engine & UI

Triple Defense Against Hallucination

1. Strict Source Constraint
Prompt mandates answers only from provided excerpts.

2. Graceful Refusal
Model states inability to answer if context is insufficient.

3. Proactive Conflict Detection
Flags and presents conflicting data from multiple sources.

Citation & Audit

Forced Citations
Every answer includes **[Source: filename, Page X]** to trace origin.

Citation Audit Module
Intercepts 97% of fake citations, ensuring data authenticity.

Immersive Gradio UI

A three-panel layout designed for seamless user experience:

- Document Management
- Chat & Evidence Display
- One-click PDF Source Jumping

Hugging Face

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03. Evaluation & Experiment Results

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03. Localized RAGAS Evaluation Loop

LLM-as-a-Judge

To ensure an impartial and robust assessment of system performance, we leverage the high-performance Qwen2.5-72B model as an automated judge. This model evaluates the quality of generated outputs against a set of predefined standards.

Core Metrics (0.0-1.0)

1. **Context Precision:** Measures how relevant the retrieved context is to the original question.
2. **Faithfulness:** Determines if the answer is factually grounded and consistent with the provided context.
3. **Answer Relevancy:** Assesses the degree to which the generated answer directly addresses the question.

Experimental Design

Gold Test Set: A curated set of 20 diverse questions, covering Factoid Extraction, Causal Reasoning, Macro Summarization, and Structured Artifact Generation.

Ablation Study: Comparative analysis across three distinct system configurations: Baseline, Advanced, and Ultimate.

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03. Experiment Results

Quantitative Results

Context Precision: Baseline (0.82) → Ultimate (0.91)
Faithfulness: Baseline (0.78) → Ultimate (0.88)

"The results show clear, measurable gains from each optimization strategy implemented."

Strategy Analysis

- **Baseline:** Struggles with macro summarization due to rigid, fixed chunking.
- **Advanced:** Significant improvement across all question types, particularly in causal reasoning.
- **Ultimate:** Establishes an absolute advantage in complex macro summarization tasks.

Verifiability Metrics

- Citation Jump Success Rate:96%(Target ⩾ 95%)
- Citation Page Accuracy:92%(Target ⩾ 90%)
- Fake Citation Interception:97%(Target ⩾ 95%)

"All metrics consistently meet or exceed preset quality assurance targets."

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04. Key Contributions

End-to-End Integration

Successfully built a complete RAG system with a closed-loop workflow, seamlessly connecting the stages from document ingestion to retrieval-augmented generation and comprehensive system evaluation.

Technical Innovations

- Adaptive Semantic Chunking & Dual-path Hybrid Retrieval
- RAPTOR Hierarchical Summarization for complex reasoning
- Page-level Citation & Traceability for answer trustworthiness

Localized Evaluation System

Established a fully offline "LLM-as-a-Judge" system for reliable performance measurement. It enables us to rigorously assess the quality of generated answers, relevance of retrieved documents, and overall system accuracy.

Complete Analyst Workflow

Delivered a user-friendly Gradio frontend that supports a seamless experience from initial Q&A to evidence verification and source exploration, fully aligning with the needs of professional analysts.

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04. Future Directions

Multimodal Capabilities
Extend the system to understand and analyze charts, graphs, and tables in financial reports.

Real-time Data Integration
Connect to live data sources like Yahoo Finance or Bloomberg for dynamic market context.

Personalized Recommendations
Intelligently recommend relevant report sections or analysis angles based on user history.

Multilingual Support
Extend support to analyze reports in Chinese, English, and other languages.

Domain Expansion
Apply the technical framework to other vertical domains like healthcare, law, and technology, broadening the scope of document intelligence applications.

Thank You for Listening!